Quarb Implementation Guide

The Quarb Project

Version: Draft 0.1 (walking skeleton)

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Chapter 1

About This Guide

This guide describes the **Quarb engine**: the Rust implementation of the Quarb query language, its `qua` command-line tool, and the adapter architecture that lets one engine query many data sources. The language itself — its data model, execution model, and syntax — is defined in the *Quarb Language Specification* (the `lang` sibling project); this guide covers only the implementation.

The guide tracks the code. The engine is being built as a series of thin, working slices rather than from a finished design. This document tracks that process: constructs the engine *already implements* are marked **[live]** and documented against the real code; constructs that are specified but *not yet built* are marked **[planned]** and sketch the intended surface. As each slice lands, its **[planned]** entries become **[live]**. The current milestone is a *walking skeleton*: rooted tree navigation over a filesystem.

Central concern: the adapter surface. The one design question this guide most needs to get right is the *adapter surface* — the trait a data source implements to be queried. That surface is deliberately being grown against a concrete first adapter (the filesystem) so that its shape is driven by real navigation problems rather than speculation. Chapter 3 is its home.

Chapter 2

Architecture

The engine follows a conventional compiler-style pipeline. A query string is lexed and parsed into a small AST, which the executor evaluates by driving an *adapter* over the target data:

```
[query string]
      |  lexer          quarb::lexer
      v
[tokens]
      |  parser         quarb::parser
      v
[query AST]             quarb::ast
      |  executor       quarb::exec
      v
[node set]  <-- drives --  [AstAdapter]  --> [data source]
```

The executor never touches the data source directly. It knows only the `AstAdapter` trait (Chapter 3), so the same query runs unchanged over any adapter.

2.1 Workspace Layout

The engine is a Cargo workspace mirroring the intended separation of concerns — the core engine, one crate per adapter, and the CLI:

Crate	Responsibility
`quarb`	The core engine, in layers (Section 2.2): `value`, `syntax`, `stdlib`, `pipeline`, `graph` (the `adapter` module defining `AstAdapter` is its). No I/O.
`quarb-fs`	The filesystem adapter. Depends on `quarb`.

`quarb-imap`	Remote IMAP mailboxes, queried in place: the maildir mapping over the wire (folders, messages by UID, headers as properties, `::::epoch`), transported by `curl` over subprocess (it speaks IMAP natively) — zero new dependencies. Lazy per message: one LIST, one UID SEARCH per folder, two fetches per touched message. The thread fabric is deliberately absent (indexing it means fetching every header — that is a sync; use mbsync + `quarb-maildir`). Target: `imap[s]://HOST[:PORT][/FOLDER]`; credentials via `QUARB_IMAP_USER`/`PASS` or `~/.netrc`. Read-only. Tested against a bundled deterministic mini IMAP server.
`quarb-json`	The JSON adapter (keys and indices as names, value types as traits). Depends on `quarb` and `serde_json`.
`quarb-gsheet`	Google Sheets on the `quarb-xlsx` mapping (sheets as tables, first-row headers, typed values via `UNFORMATTED_VALUE`), fetched from the Sheets API. Auth: the API rejects gcloud's `cloud-platform` tokens, so the friction-free path is an API key (`?key=` / `QUARB_GSHEET_KEY`; enough for link-readable sheets), with a properly-scoped bearer for private ones. Target: `gsheet://SPREADSHEET_ID`.
`quarb-html`	The HTML adapter (tags as names, block/inline as traits, attributes as properties, anchor fragments resolve). Depends on `quarb` and `scraper`.
`quarb-pdf`	PDF internals as the object graph: indirect objects are nodes named by their declared `/Type`, scalar dictionary entries are properties, references are edges labeled by their key (`->Kids`, `<-Parent`; array references hoist to the array's owner), with the reverse index built at load. Depends on `quarb` and `lopdf`.

`quarb-text`	The shared text-level model — the relational pattern for documents: the per-format crates lower their format into one block-event vocabulary (the atrep shape: kind, taxis, lemma, body, hypograph), and this crate alone implements the adapter. Sections with explicit title (`::lemma`) + body are derived from the flat heading stream by the outline rule; paragraphs, blockquotes (attribution as `::hypograph`), lists, and verbatim blocks (`::::lang`) complete the vocabulary; tables denormalize into nested lists carrying the `<table>` trait. Bare `::` is the flattened prose of the subtree. The apparatus is first class (litogramma's model): `footnote`/`endnote` callouts and bodies share their family name, resolve by (family, onym) into `->footnote`/`->endnote` edges, and carry `<deixis>` (the callout), `<note>` (the body), and `<dangling>` traits; the `aside` family (anchored margin content) rides the same machinery without `<note>`, and margin-spelled footnotes carry `::::form`; `index-mark` nodes expose declared index marks (`::term`) invisible to the prose. Plain text (`.txt`, blank-line paragraphs) mounts here by default; `text:` forces the reading for html/md. Depends on `quarb`.
`quarb-text-docx`	The Word text-level adapter, from declared structure only: outline levels resolved through the style chain (never style names), the accepted view of tracked changes, `w:tblHeader` header rows, the closed cited quote-style set, both note parts (`word/footnotes.xml`, `word/endnotes.xml`), and `XE` field instructions as index marks. Depends on `quarb-text`, `zip`, and `quick-xml`.

`quarb-text-odt`	The OpenDocument text-level adapter (`.odt`, the flat `.fodt`, OpenOffice.org's `.sxw`), the Word reader's sibling: `text:h` outline levels and a paragraph style's `default-outline-level` along the parent chain, the declared list containers with their style's level kind, `table-header-rows`, the same cited quote style, inline `text:note` bodies lifted to the document end with the callout at its offset, bookmarks and reference marks as bearers or anchors, `text:a` and the reference fields as refs, index marks, emphasis and code spans from character styles' declared properties, the accepted view of tracked changes, `meta.xml` as the declared identity. Depends on `quarb-text`, `zip`, and `quick-xml`.
`quarb-text-rtf`	The Rich Text Format text-level adapter, from declared control words: `\outlinelevel` direct or through the stylesheet's `\s` / `\sbasedon` chain, the list table's level kinds behind `\ls` / `\ilvl` (and the old `\pn` group), `\trhdr` header rows, the same cited quote-style names, inline `\footnote` groups as callouts with their bodies at the end (`\ftnalt` the endnote), HYPERLINK fields as refs, bookmarks, `\xe` index marks, `\deleted` dropped, emphasis and code spans from the run's declared formatting, the code page of `\ansicpg` for the `\'xx` bytes (reported as the source's declaration), the info group as the declared identity. Its own tokenizer; depends on `quarb-text` and `encoding_rs`.

`quarb-text-doc`	The legacy Word (97–2003, OLE2) text-level adapter, the text-only first cut: the `WordDocument` stream's FIB names the table stream and the CLX, the piece table gives the text (8-bit pieces in the FIB language's code page, UTF-16 pieces as they are), paragraph and cell marks split paragraphs, a field's instruction is dropped and its result kept, the footnote text's bodies are split by `PlcffndTxt` with the footnote marks as their callouts, and `SummaryInformation` declares the identity. No outline levels, lists or tables yet (they need the property pages and the stylesheet). A plain-text file named `.doc` is told apart by the OLE2 magic. Depends on `quarb-text`, `cfb` and `encoding_rs`.
`quarb-text-epub`	The EPUB text-level adapter: container to rootfile to spine, chapters through the HTML adapter in spine order (`linear="no"` excluded); the EPUB note vocabulary arrives via `quarb-text-html`. Each chapter's ids are its own namespace (`ch2.xhtml#n1`): a reused `id` names one bearer per chapter, and a reference lands in its own chapter or the one it names. Depends on `quarb-text`, `quarb-text-html`, and `zip`.

`quarb-text-html`	The HTML text-level adapter: soup dropped (nav, chrome, scripts, forms, media), the head read for the page's declared identity (`title`, `lang`, canonical, every `meta`, `rel=tag` links, a JSON-LD block — verbatim as `::::` metadata on the document, and curated into tags-as-traits, `::category`, `::description`, `::author`, `::published`; every node answers `::href` / `::anchor` / `::link`, and `| link` rides them), wrappers transparent, headings flat into the shared derivation, trailing `cite`/`footer`/`figcaption` as quote attribution; notes only from the declared vocabularies (`epub:type` noteref/footnote/endnote/rearnote/marginalia, DPUB-ARIA `role` doc-*), endnote words landing in the endnote family, marginalia in the aside family. Depends on `quarb-text` and `scraper`.

`quarb-text-koine` The koine adapter — the hub's first rung: a koine/litogramma document (parsed by the `atrep` crate, settled by `kanonizo::tasso`) lowers into the shared block vocabulary through its dialektos-resolved sim names — enclosing headings by nesting depth, both note families and asides by name, index endos as marks, the table model (`table` / `row` / `header-row` / `cell`, the `hspan` / `vspan` monosims padding the cells a span covers, so every row is a record of the same width) and the older stichoi tables, refs keeping their key as written. The dictionary vocabulary (spec: The Dictionary Vocabulary): a lexigramma document lowers its `entry` and `sense` sims to containers of those kinds (the headword the lemma and the onym, the taxis the homograph or sense number), the inline apparatus to spans named by their sims with `::kind` / `::lang` from the genos, an `inflection` to a `<hidden>` node outside the prose, the languages line to `::::languages`, and the definition list typed `.abbreviations` (or, left untyped, the one before the first entry) to the abbreviations list (`::kind`) the labels resolve against (`-->`). Foreign formats arrive through atrep's endomorphosis importers (`atrep::endo`) into their mirror dialects — Markdown, HTML, reStructuredText, Org, djot — read by the same name-keyed mapping (flat `heading-N` endos, flat item runs wrapped, rst footnotes into the shared apparatus); Word through `atrep::docx` (`koine:report.docx`, declared structure only, beside the native `text:` reading); a Lingvo DSL dictionary (`.dsl`, `.dsl.dz`) through `atrep::dsl` into lexigramma, the `#INCLUDE`d abbreviations read beside it, under `koine:`, `text:` and `corpus:` alike (`lit:` refuses: a dictionary is no literary edition). Serves `koine:` (and `text:` on `.atd`/`.atk`, where the routes converge); `.xml` dispatches by declared identity (root namespace, DOCTYPE public id, unambiguous root element) into the TEI/DocBook/JATS/USX/OSIS importers, `?format=` forcing any format; a bare `<article>` and everything nothing imports yet — LaTeX included — refuse naming the working spelling. Depends on `quarb-text` and `atrep`.

quarb-text-latex	The LaTeX text-level adapter: the declared structural subset of a .tex source — declared sectioning commands, footnotes (the at-callout convention normalized to document-end bodies), the endnotes package on its own counter, Tufte sidenotes into the footnote family (::::form), \marginpar / \marginnote as asides, \index marks, quote/quotation, itemize/enumerate, verbatim; prose by drop list and unwrap-by-default, definition argument trains consumed whole. Depends on quarb-text.
quarb-text-markdown	The Markdown text-level adapter, mapping pulldown-cmark events directly (no HTML round-trip): heading levels, list ordinals, fence languages, and pipe tables come from the source. Depends on quarb-text and pulldown-cmark.
quarb-text-pdf	The PDF text-level adapter — the reader's model from the two declared rungs: the /Outlines tree (titles as ::lemma, destinations bounding each section's extent) and exact-baseline line geometry (bit-for-bit equal y, ::::page/::::y metadata; ToUnicode CMaps decoded, one stated kern constant for word gaps). No paragraph is ever minted. Depends on quarb-text, quarb-pdf, and lopdf.
quarb-text-djvu	The DjVu text-level adapter (text:scan.djvu), by the PDF reading's doctrine: the NAVM outline is the heading ladder (sections holding their pages, front matter at the root; page/* without one); a page's zones are its tree as the text layer names them (column, region, paragraph, line; words a line's text; declared paragraphs kept, none minted); ::::page/x/y/ width/height on every zone; ::::text-layer on page and document; the ANT metadata as the document node's properties. Bundled, single-page and indirect documents; the text layer only (no OCR). The plain mount of a .djvu names the reading. Depends on quarb and djvu-rs (pure Rust, MIT).

quarb-text-ofd	The OFD text-level adapter (text:invoice.ofd; GB/T 33190, China's fixed-layout format), by the PDF reading's doctrine: the Outlines tree is the heading ladder (a section's extent from its destination — a page and a top, or a named bookmark — to the next entry's; front matter at the root; every listed page as page/* without one); a line is the text codes a page shows at one declared baseline in x order (a gap between two codes' extents a space; no paragraph minted); ::::page, the bounding box ::::x/y/width/height in the page's millimetres, ::::baseline; a page's template pages join its lines; DocInfo (title, author, subject, abstract, keywords, dates, creator, custom data) as the document node's properties. Attachments and signatures read under the plain mount, which opens the package as a zip. Depends on quarb, quick-xml and zip.
quarb-text-hwpx	The Hangul Word Processor text-level adapter (text:report.hwpx; KS X 6101, OWPML — the zip of XML Hancom's office writes), in the Word and OpenDocument readings' vocabulary from declared structure alone: a heading is a paragraph whose paragraph property declares an outline level (style names never matched), a list the run of paragraphs declaring a numbering or bullet, nested by level; tables through the shared table rule (header="1" cells the header, column spans padded); foot- and endnotes as callout and document-end body; hyperlink fields as refs, bookmark fields as anchors; tracked changes as the accepted view; headers, footers, pictures, drawings and equations not carried; the package file's metadata as the identity. The binary HWP 5 is not read yet; the plain mount of an .hwpx is the zip of its XML. Depends on quarb, quarb-text, quick-xml and zip.

`quarb-text-uof`	The Uniform Office Format text-level adapter (`text:report.uot`; GB/T 20916, UOF 1.0 one XML, UOF 2.0 a zip package, both by the root `uof:UOF`), in the Word and OpenDocument readings' vocabulary from declared structure alone, the Chinese element names matched as written: a heading is a paragraph declaring an outline level (on itself or through its paragraph style), a list the run of paragraphs naming an automatic numbering (a bullet at its first level unordered), nested by the declared level; tables with their declared header rows; foot- and endnotes as callout and document-end body; hyperlink regions resolved through the link set, bookmark regions through the bookmark set; tracked changes as the accepted view; anchored objects, comments and formatting not carried; the metadata as the identity. The spreadsheet and presentation modules are not read. Depends on `quarb`, `quarb-text`, `quick-xml` and `zip`.
`quarb-xml`	The XML adapter (qualified tags as names, attributes as properties, `id`/`xml:id` references resolve; strict well-formedness, no traits). Depends on `quarb` and `quick-xml`.

`quarb-relational`	The shared relational model: tables as root children, rows named by primary key, columns as properties, declared foreign keys driving `-->` resolution (chained hops instead of JOINs), `->`/`<-` crosslinks labeled by column, `<--(::fk)` reverse resolution from the declared keys (`reverse_resolve` on the relational model), and the table-naming reference label for undeclared references (required where a column declares no key). Loading is catalog- eager, rows-lazy, hops by key: a table is read whole only when a query lists it (`/t/*`, `::::n-rows`); a hop reads the rows it lands on through the driver's keyed source (`WHERE key IN (…)`, one statement per frontier; such a row carries no row id until the table is read, never a sentinel value a real row could bear), rows keep their identity when the table is later read whole, and untouched tables are never read (`::::loaded`, `::::n-loaded` report the state). Engine adapters are thin catalog drivers over it. Depends on `quarb`.
`quarb-sqlite`	The SQLite catalog driver: PRAGMA introspection, storage-class type mapping. Depends on `quarb-relational` and `rusqlite`.
`quarb-objstore`	GCS and S3 buckets as lazy directory trees (`gs://` / `s3://`, delimiter listings, paginated; object content fetched on first read). Composed by default in `qua`: a bucket of JSON / CSV / source files grafts, so `/data/store.json/books/*` crosses from cloud storage into the document mid-path — grafting is this adapter's point. GCS: anonymous for public buckets, gcloud-token auth otherwise. S3: anonymous only in v1 (SigV4 signing is a recorded extension — private buckets refuse).
`quarb-postgres`	The PostgreSQL catalog driver: `information_schema` introspection over `tokio-postgres` (internal current-thread runtime, synchronous surface); native bool/int/float/text, everything else `::text`-cast into the numeric-reading regime. Depends on `quarb-relational`.

`quarb-mysql`	The MySQL/MariaDB catalog driver: `information_schema` introspection over `sqlx` (internal current-thread runtime, synchronous surface; `key_column_usage` carries the foreign keys directly); integer/float/character families native, everything else `CAST(... AS CHAR)` into the numeric-reading regime. Depends on `quarb-relational`.
`quarb-mssql`	The SQL Server catalog driver: `INFORMATION_SCHEMA` plus `sys.foreign_key_columns` (parallel column ids keep composite keys paired), over `tiberius` (pure-Rust TDS, internal current-thread runtime); `bit`/integer/float/character families native, everything else `CAST(... AS NVARCHAR(MAX))`. Target: `mssql://USER:PASS@HOST[:PORT]/DATABASE`. Depends on `quarb-relational`.
`quarb-oracle`	The Oracle catalog driver: the `USER_*` dictionary views over the `oracle` crate (ODPI-C, synchronous); `NUMBER(p,0)` integer, `BINARY_FLOAT`/`DOUBLE` float, character family text, `DATE`/`TIMESTAMP` through an ISO-8601 `TO_CHAR` into the temporal regime, everything else `TO_CHAR`. Target: `oracle://USER:PASS@HOST[:PORT]/SERVICE`. Depends on `quarb-relational`.
`quarb-ldap`	The LDAP directory driver: the DIT *is* the arbor — the base DN is the root, an entry's children are the entries one level below (RDN names), its attributes are properties (multi-valued as lists), `objectClass` values are traits. DN-valued attributes (`member`, `manager`, ...) are native references — `~>` follows, `->` enumerates, `<-member` finds the groups by a filtered search. Over the `ldap3` sync client, anonymous or simple bind. Target: `ldap[s]://[USER:PASS@]HOST[:PORT]/BASE_DN`.

`quarb-cbor`	The CBOR (RFC 8949) adapter, the JSON mapping extended: non-text map keys name nodes by canonical rendering, byte strings are hex scalars (`::::length` is the raw byte count), the standard time tags (0, 1) decode to instants, other tags are transparent with the number at `::::tag`. Traits carry the CBOR type. Binary, by the `.cbor` extension. Depends on `ciborium`.
`quarb-csv`	The CSV/TSV adapter (one `row` node per record, columns as properties, empty cells as missing values that numeric aggregates skip; `::::columns` / `::::n-rows` metadata). Depends on `quarb` and `csv`.
`quarb-xpath`	The XPath 1.0 bridge, both directions: `translate` imports an XPath expression to Quarb, and `export` walks a Quarb query's reflection arbor (the locked vocabulary as the tooling surface) back to XPath — explicit refusals and divergence notes both ways. Binaries `xpath2quarb` / `quarb2xpath`; the export side is differentially tested against `xmllint`.
`quarb-jq`	The jq bridge, both directions: `translate` imports a jq filter (value-projecting; a node mode pairs with `jq 'path(...)'`), and `export` walks the reflection arbor back to a jq filter. Binaries `jq2quarb` / `quarb2jq`; the export side is differentially tested against the `jq` binary.
`quarb-bigquery`	The Google BigQuery driver: the REST API directly (`jobs.query` over a synchronous HTTP client — no cloud SDK, no async runtime), bearer token from `QUARB_BQ_TOKEN` or `gcloud auth print-access-token`, catalog from `INFORMATION_SCHEMA` including the *unenforced* key constraints (declare them and the reference machinery works). The execution ladder is a billing instrument here: BigQuery meters bytes scanned, so lazy loading, pushdown, and `--save` directly reduce cost. Target syntax: `bigquery://PROJECT/DATASET[?account=EMAIL]`.

`quarb-duckdb`	DuckDB, the fifth catalog driver — and the columnar side door: a `CREATE VIEW ... AS SELECT * FROM read_parquet(...)` makes Parquet first-class. Keys from `duckdb_constraints()` (the `information_schema` FK view reports the referencing side only), rows lazy, `raw_query` for the ladder, read-only connection. Bundled build; `.duckdb` / `.ddb` by extension.
`quarb-xlsx`	Spreadsheets (`.xlsx` / `.xls` / `.ods`, via calamine) on the relational model: sheets as tables, first-row headers as columns (positional `cN` for blank/duplicate), rows named by sheet row number, numeric cells typed. No keys — sheets declare none — so resolution needs hints. Dispatched by extension *before* the archive check (workbooks are zips; the sheets are the point).
`quarb-firebase`	The Firebase Realtime Database driver: the RTDB *is* a JSON tree, so the mapping is `quarb-json`'s, fetched lazily one node per shallow GET and cached. Properties are direct fetches (`::score` is one tiny request); `~>` resolves by hint naming a root-relative container (`::parent~>item`). Containers that refuse enumeration (permission-scoped or unbounded — the public Hacker News database's `/v0/item` holds tens of millions of keys) are *opaque*: reachable by name, empty to wildcards — served by the engine's new `children_named` fast path. References are *declared client-side*: a refs document (`{"refs": {"parent": "item", "kids/*": "item"}}`, supplied via `qua --refs`) is the portable mini-catalog a schemaless substrate cannot hold — with it, bare `~>` resolves and `->` crosslinks enumerate (array fields one probed edge per element); an inline hint overrides. Reverse resolution stays empty (it would scan what opaque containers refuse; server-side `.indexOn` queries are the recorded v2 path). Target: `firebase://HOST/PATH[?auth=...]`. Read-only.

`quarb-tree-sitter`	Source code as its literal syntax tree — the syntax level (né `quarb-code`): node kinds as edge names (`//function_item`), tree-sitter *fields* as properties (`::name` on a function is its name), source text as the value, `::::start-line` / `::::kind` / `::::field` metadata; named nodes only. Rust / Python / JavaScript / C by extension, compile-time grammar set. Grafts under composition — `--graft` queries every function in a directory tree at once.
`quarb-code`	Source code at the code level: declaration identifiers are node names (`/lexer/lex`), anonymous constructs keep a normalized cross-language vocabulary (`if`, `switch`, `for`, `call`), everything else dissolves (children hoist). Curated traits (`<function>`, `<type>`, `<loop>`); `::signature` / `::doc` / `::callee`; raw backend kind only at `::::kind` (aliased, closed surface); `->definition` file-local crosslinks, one declaration list per identifier: a qualified callee (`Type::new`, `obj.helper`) prefers the declarations under that qualifier, an unqualified one fans out. Opt-in via the `code:` target prefix; `code:DIR` grafts every supported source leaf — filesystem into declarations, one namespace.
`quarb-compose`	Adapter *composition*: wrap any adapter and a leaf whose name or content parses (json / xml / html / csv, else a sniff) grafts its parsed arbor as children — `/data/store.json/books/*` crosses from filesystem into JSON mid-path. Marked-up members graft at the DOM by default or, under `DocumentGraft::Text` (`text:` on an archive), as the text level; the source-level grafts (tree-sitter, the code model) sit behind the default `source` feature so the wasm build can leave them out. Grafts are lazy (read + parse on first entry), the inner root is identified with the outer leaf (no phantom node), and locators mark the boundary jar-style (`store.json!/books/1`). `qua`: default for archives, `--graft` for directories.

`quarb-archive`	Zip, tar and 7z archives as directory trees (entries as nodes, content as value, `::::size` / `::::compressed`), every format through a pure-Rust library so the wasm build reads them too: zip members deflate, deflate64, bzip2 (`bzip2` with `libbz2-rs-sys`), lzma (`lzma-rs`); tar plain and under gzip (`flate2`), xz (`lzma-rs`), bzip2, zstd (`ruzstd`); 7z (`sevenz-rust`, LZMA/LZMA2). The format is told by its magic bytes. Zip members in zstd or xz and rar would need a C library and are not read by default: rar rides the opt-in `rar` feature over the `unrar` crate (rarlab's library under its own licence), on for `qua`, off for the playground and the extension, and a RAR without it is refused naming the feature. CHM help files read through `libchm` (pure Rust, `ITSF`): members under their paths, the system records left out, a page without a `<meta charset>` decoding under the `#SYSTEM` locale's code page. An MHTML web archive (`.mhtml`, `.mht`, or MIME header lines) reads through an in-crate RFC 2045/2046 splitter: each part a member under its `Content-Location` path, transfer-decoded, its charset the declaration. Members decode under the mount's source-boundary rule.

`quarb-web` The *web level*: a site as an arbor — `/sites/<host>/pages/<path>` for every page (directories from the URL paths, `<dir>` / `<page>` / `<redirect>`), the declared categories and tags as rows of their own containers, every hyperlink of the DOM resolved into `->link` / `<-link` (`->category`, `->tag`, `<-page`, `<-category`), the degrees, `::pagerank`, `::redlinks`, and `<orphan>` as columns; each page grafts at the text level on entry, and a prose `ref` that leaves its page resolves to the page it names. Rows answer to their kind (`//page`), their segment, its stem, their title. A `WebStore` contract behind it; stores: in-memory from a tarball (`archive`, the wasm `Site`) or a directory (`fs`), SQLite (`sqlite`), PostgreSQL (`postgres`; COPY export, deferred foreign keys), one schema in two dialects. A planner over the reflected query answers on three rungs — Full (a bare count, every conjunct exact), Prefilter (a superset of the enumerating step's matches; the engine re-runs the query over the level scoped to the candidates), Scan — with the text prefilter resting on `page_text.plain` = the graft's own `plain_text()`. The four adapter hooks (`document_by_ref`, `reverse_resolve`, `descendants_named`, `prefetch_links`) let a store answer where the engine would walk. `quarb-web-ingest` (`cli`): `crawl` a directory or archive into a store, `wikipedia` the Wikimedia Enterprise HTML dumps (streamed; links by URL then through redirects; `--tree=path|category`), `analyze`, `index` (pg_trgm). `qua`: `web:<dir|archive|site.db|postgres://…>`, `--explain` / `--no-pushdown`.

`quarb-trace`	The `quarb/trace` viz contract as Rust: the payload types (arbor snapshot, steps, threads, snapshots, window and stubs, positions, ribbons, lookahead verdicts, results) and their kaiv writer; no dependencies, no reader. `quarb::trace::trace` fills one by *replaying* the query rather than instrumenting the evaluator: one hop per live thread through the evaluator's own `navigate_paths` (candidates with the predicates stripped, survivors with them — the fork, the verdicts, the walked edges), `apply_command` per thread for filters and over the whole context for aggregates, a hand-iterated walk for quantified groups (one frame per depth), then checks the final context against a plain evaluation. A staged suffix marks the steps beyond the committed query (a filter preview); a window selects the slice of a large result list to embed and what the stubs on either side measure. `Doc::trace` in `quarb-session` is the entry; the renderers read the kaiv document (quarbopsis: `quao trace`).
`quarb-maildir`	Maildir (`cur/`+`new/`) and mbox mailboxes: headers as properties (lowercased), the body as the value, `::::epoch` parsing the Date header for range predicates. The thread graph is reference fabric: `::in-reply-to~>` walks to the parent message, `<-in-reply-to` finds the replies, via a Message-ID index. Never touches flags; read-only.
`quarb-firestore`	Google Firestore (native mode): collections and documents alternate as the arbor's levels; map and array fields descend as children, subcollections beside them. `referenceValue` fields are *native* references — the first remote adapter where `~>` and `->` need neither a schema nor a refs file. Lazy per level, REST direct, token via `QUARB_GCP_TOKEN` / gcloud. Target: `firestore://PROJECT/DATABASE[?account=...]`.

`quarb-datastore`	Google Cloud Datastore (Firestore in Datastore mode): kinds and entities, the most relational of the document stores; `keyValue` properties are native references (`~>`/`->` schema-free). Kind list via a `__kind__` query, entities via cursor-paginated `runQuery`, lazy per kind. Target: `datastore://PROJECT[/DATABASE][?account=...]`. Both drivers send the `x-goog-request-params` routing header named databases require.
`quarb-mongodb`	MongoDB: collections hold BSON documents, named by `_id` (ObjectIds as hex); embedded documents and arrays descend as children. DBRef fields (`{$ref, $id}`) are native references (`~>`/`->` schema-free). BSON dates arrive as instants; ObjectId and Decimal128 as strings. Lazy per collection (one `_id`-sorted `find`), reference targets by single `findOne`s; official Rust driver, sync API. Read-only. Target: a standard connection string with the database as the path, `mongodb://HOST[:PORT]/DATABASE[?options]` (or `mongodb+srv://`); a missing `serverSelectionTimeoutMS` defaults to 5 s.
`quarb-kubernetes`	The Kubernetes API, which is already a tree: the root holds one listing per discovered resource (cluster-scoped kinds plus the all-namespaces view, kubectl's `-A`); each namespace object carries its scoped listings beside its own fields; objects descend as their JSON, RFC 3339 strings as instants. `ownerReferences` are native references (`::owner~>` climbs Pod → ReplicaSet → Deployment schema-free; `->node`, `::namespace~>`); unknown property names fall through to `metadata.labels`. Lazy per listing, discovery at connect. Transport is `kubectl get --raw` over subprocess (git-adapter posture, zero new dependencies; `QUARB_KUBECTL` overrides the binary), so contexts and auth behave exactly as kubectl does. Read-only. Target: `k8s:` / `k8s:CONTEXT` (alias `kubernetes:`).

`quarb-gcplogs`	Google Cloud Logging as a bounded snapshot — never a tail: the target must carry `since=` (a gcloud freshness duration or an ISO instant) and/or `limit=`, and an unbounded mount is refused. Entries oldest-first at the root as `/entry` children (the `/row` convention); severity as text property, lowercased trait (`/entry<error>`), and ordered numeric rank (`::::severity`, GCP's enum values); timestamps as typed instants; `::trace` the bare trace id — the `<=>` join key across services. Unknown property names fall through payload top level → labels → resource labels → httpRequest. `filter=` passes a raw Cloud Logging filter to the server (the mount-side cut); `until=` joins it as a timestamp clause. Transport is `gcloud logging read --format=json` over subprocess (kubectl posture; `QUARB_GCLOUD` overrides — the bottled test fixture's hook; `account=` picks the gcloud account). Read-only. Target: `gcl:PROJECT?since=1h&filter=…&limit=N` (alias `gcplogs:`).
`quarb-cwlogs`	AWS CloudWatch Logs, the gcplogs design sheet over SigV4 (shared `quarb-aws` signing, credential chain, region resolution): bounded snapshots (`since=`/`limit=` mandatory), log groups as short-named root children with lazily fetched `/entry` events (or one group's entries at the root when the target names it), JSON messages decoded with property fallthrough (`::level`, `::requestId` — the cross-group join key), `filter=` as a server-side CloudWatch filter pattern, `::::group`/`::::stream`/`::::received`/`::::id` metadata. `endpoint=` overrides the URL (LocalStack; the bottled `cwlogs-server.py` fixture). Read-only. Target: `cwl:[GROUP]?since=1h&filter=…&limit=N` (alias `cloudwatch:`).

`quarb-azlogs`	Azure Monitor Logs through the Log Analytics KQL endpoint — the adapter composes the KQL itself (bounded table scan, `filter=` as a `| where` clause, `take` from `limit=`); `table=` and a bound are mandatory. One table's rows at the root as `/entry`, a comma list as root children per table. Columns are properties (`datetime` as instants, `dynamic` decoded into subtrees with fallthrough); `::OperationId` is the correlation key; the default value is `Message` or the `Name`+ `ResultCode` request line. Bearer token from `$AZURE_LOG_TOKEN` or the az CLI (`QUARB_AZ`); `endpoint=` overrides (the bottled `azlogs-server.py`). Read-only. Target: `azl:WORKSPACE?table=…&since=1h&limit=N` (alias `azlogs:`).
`quarb-cflogs`	Cloudflare edge logs, Logpull-shaped: `GET zones/{zone}/logs/received` NDJSON, one flat record per request, fields as properties under their own names plus minted `::ray` and `::timestamp`; the default value is the request line; `::::cache` the cache status. `since=` is mandatory (Logpull demands a range; `until` clamps a minute behind now). The Ray ID is what the origin sees as `cf-ray`, so mounting the edge next to origin logs makes request reconstruction cross-provider — one `<=>` on the ray, another on the origin's own trace. `$CLOUDFLARE_API_TOKEN`; `endpoint=` replays Logpush archives (the bottled `cflogs-server.py`). Read-only. Target: `cfl:ZONE_ID?since=1h&limit=N` (alias `cflogs:`).

quarb-ddlogs	Datadog Logs through the v2 search API (POST /api/v2/logs/events/search, cursor-paginated): bounded snapshots (since=/limit= mandatory), query= as Datadog's own search syntax run server-side. Severity is Datadog's ::status plus the family trait and ordered ::::severity rank; ::trace is minted from dd.trace_id; tags split into fallthrough properties (env:prod answers ::env); custom attributes navigate as children. Auth $DD_API_KEY/$DD_APP_KEY, site from site=/$DD_SITE; endpoint= overrides (the bottled ddlogs-server.py). Read-only. Target: ddl:?since=1h&query=…&limit=N (alias datadog:).
quarb-github	GitHub, three shapes in their native registers. The tree mirrors github.com URLs: users/orgs at the root (addressed by login — the root does not enumerate), repositories as their children, files / issues / pulls / releases sections below; file content and issue bodies are node values. The social graph is edge fabric, API-backed both ways: follows, starred, owner, parent (<-parent = the forks), author/assignee, org, and contributor — the first adapter with true backlinks, and its contributor edge carries data ($-::contributions). Counts and scalars are properties (::stars, timestamps as instants); flags, issue labels, and repo topics are traits (<fork>, <bug>, <cli>); an organization answers the <org> shorthand and the forge-neutral <group> alias. Direct addressing skips enumeration (/rust-lang/rust/issues/12345, closed included; listings show the API's default open set). Transport is gh api over subprocess (QUARB_GH overrides); lazy, cached, read-only. Target: github:[OWNER[/REPO]].

`quarb-gitlab` GitLab, whose defining shape is the *deep tree*: groups nest arbitrarily and mix subgroups with projects as children, which the REST API walks by URL-encoded full paths and the arbor restores to a path — `/tesslab/instruments/gauge/issues/7`, one GET per segment. Project sections: `files` (content as value), `issues` / `mrs` (the open sets; direct addressing by iid reaches any state), `releases`, `pipelines` (jobs as children; statuses as traits — CI is half of what a GitLab project is). Code over social: no follows, stars as a count; the edge fabric is `member` (with access-level edge data — `$-::access`, `$-::role`), `parent` on forks (`<-parent` = the forks), and `author` / `assignee` / `reviewer`. Labels (scoped included), topics, visibility, and statuses are traits; a project answers the forge-neutral `<repo>` alias (`//*<repo>` sweeps either forge identically). Transport is `glab api` over subprocess (`QUARB_GLAB` overrides; hosts via glab config), so self-managed instances work as glab does. Read-only. Target: `gitlab:[PATH]`.

`quarb-git` The git repository driver: the object DAG as the arbor. `/branches`, `/tags`, `/HEAD` (refs alias their commits — same properties, same children), `/commits` (rev-syntax aliasing: `/commits/'HEAD~2'` navigates without enumeration; `/commits/*` batches one `rev-list --all`). A commit's children are its *tree* — time-travel file access, blob content as the node value. References are inherent: `::parent~>`, `->parent` (merges fan out), `<-parent` (children, from the enumeration). The diff surface: tree entries carry the `<changed>` trait when their path is in the commit's diff vs its first parent (trees by prefix), so `git log -- path` is `/commits/*[/path<changed>]`; commits answer `::changed` (the touched paths, deletions included) and `::::n-changed`. Plumbing over subprocess — no libgit2, zero new dependencies. Target: `git:PATH`. Read-only.

`quarb-atrep`	The atrep semantic-document adapter — the OQL homecoming. Mounts a deltos (`.atd`) or kanon (`.atk`) *through its dialektos*: sims named by their dialect names (`//section`, `//footnote`), genoses as traits, `::onym`/`::taxis`/ `::lemma` projections, deixes and onym-monosims as typed crosslinks with a precomputed reverse index (the first adapter to prebuild both `links` and `backlinks`), `<dangling>` as a broken-cross-reference linter. Mounting runs `kanonizo::tasso` (ordering only — onyms stay as authored; transclusion stays visible as edges). Depends on `quarb` and `atrep`.
`quarb-serve`	The *serve protocol* — the extension surface for tools the engine never links: a tool implements `AstAdapter` over its own model and hands it to `serve()` (three lines; non-Rust tools implement the protocol directly — nine operations over line-oriented JSON), and `qua serve:COMMAND` spawns it and mounts it beside every other source. LSP's shape: the protocol is the plugin system. Two wires, one protocol: the handshake is JSON (the universal floor) and advertises formats; every first-party server offers *kaiv* (`.daiv` frames, natively typed leaves under `/@serve`) and the client upgrades to it — the house format is the default wire, JSON the foreign-tool floor. Read-only by construction (no mutating operation exists). First external user: `cuj quarb-serve` (the cuj vault, mounted without qua knowing cuj). Reference server: `quarb-serve-json`; identity-tested against the direct adapter.

`quarb-sql`	The SQL bridge, both directions: `translate` imports a `SELECT` statement (WHERE, one JOIN via correlation and the witness, GROUP BY/HAVING, ORDER/LIMIT/DISTINCT, `LIKE` as case-insensitive regexes), and `export` walks a Quarb query's reflection arbor back to `SELECT` — witness joins become `JOIN ... ON`, grouped reductions become `GROUP BY`/`HAVING`; `~>` chains are refused by design (foreign keys live in the schema, not the query — spell `<=>` to export). Both directions differentially tested against SQLite; binaries `sql2quarb` / `quarb2sql`.
`qm`	The mailbox interface for the terminal — the first *library-path* client beside `qua`: it opens the Maildir adapter once per invocation, addresses messages as node handles, and phrases its subcommands (`ls`, `read`, `thread`, `from`, `grep`, `count`) as queries where a query is the natural shape, with `qm query` as the raw escape hatch. Actions are the tool's, not the engine's: `qm seen` performs the Maildir flag rename itself on the file handle the adapter exposes (`path_of`).

`qua`	The command-line tool. Picks the adapter from its argument (a directory is filesystem; `.csv`/`.tsv` is a table; a file or piped stdin is XML by extension or a leading `<?xml`, HTML by extension or a leading `<`, else JSON) and prints results. `--xpath` / `--jq` / `--sql` translate the query from XPath, jq, or SQL first; database inputs get *pushdown* — a query whose constructs are all in the verified-safe set (filtered selections, whole-table aggregates, witness joins) compiles to SQL and runs on the server, with rows ordered by the catalog key so output is identical to the scan path (`--no-pushdown` opts out; any refusal or error falls back silently, and `--explain` prints the decision — the pushed SQL, or the construct that kept the query on the scan path). When full pushdown refuses, *partial* pushdown pushes the leading row predicates as the table's fetch filter and runs the original query over the reduced set (gated: single reach, no crosslink/resolution axes, no metadata — anything that could observe the filtering). `--save FILE` materializes a result — `.db`/`.sqlite` as a table (`--as` names it), `.json` as a JSON array — both first-class inputs for later queries (the reference syntax is the existing file/mount mechanism); `--daiv` emits canonical kaiv (typed leaves with per-value provenance; needs the sibling `kaiv-rs` checkout).

Adapters are separate crates by design: `quarb-fs`, `quarb-json`, `quarb-html`, and `quarb-xml` share nothing but the `AstAdapter` trait, and a consumer pulls in only the adapters it needs. Each new adapter has confirmed the surface without changing it — the filesystem, a JSON tree, an HTML DOM, and an XML document are genuinely the same arbor to the engine.

2.2 The Layers of the Engine Crate

The `quarb` crate is one crate in layers, each a directory under `quarb/src/`, each allowed to reference only the layers below it:

Layer	Holds
`value`	The value space: null, bool, int, float, text, list, record, table, foreign; numbers, quantities, locales, transliteration, encodings. What an AST literal holds. No node.
`syntax`	The language as text and tree: lexer, parser, AST, unparser, highlighter, completion, the function registry. One grammar for the calculator and the graph language.
`stdlib`	The functions over values, reached by name from a call, with the arithmetic and comparisons the operators run.
`thread`	The forking machine: a list of threads and three moves (fork, terminate, the whole-list map); every layer above hangs its own per-thread state on the thread, copied on a fork and never shared afterwards.
`pipeline`	Commands, control operators, bodies, environments, the registry and memo of a run, provenance.
`graph`	Arbors, anchors, steps, correlation, adapters, identity, tabula reading, reflection, the trace. Everything that knows a node.

`lib.rs` is the facade above all of them: the public paths (`quarb::adapter`, `quarb::Value`, `quarb::run`, ...) are re-exports and did not move. The dependency direction is checked, not assumed: `scripts/layers.py` resolves every `crate::` path in a layer's files to its layer and fails on a reference upward that `scripts/layers-debt.txt` does not record. The record is the debt still being paid and is meant to burn down to nothing; a change that adds an upward reference fails the check, and one that removes some must update the record in the same commit. Six references remain: the parser's default reflector (a test convenience), the library's two kaiv arms, and the graph tests' reach into the facade.

The thread and its attachment. The thread (`thread/mod.rs`) is generic over what the layer above hangs on it: `Thread<X>` carries a register, a topic with its provenance, the members of a group, the captures of the last match, and one `ext: X`. The `Attachment` trait names what a scope may see of the attachment beside the attachment itself: a `View` (the graph's: whether the thread stands on a node), a `Crossing` (the edge a hop is crossing, for `$-`), a `Witness` (what a correlation search tries in place of the thread's own bindings), and a `Key` (what a context is indexed by). The calculator's attachment is the unit type, so a calculator thread is the register and the topic and nothing else. The graph's is `GraphExt` (`graph/navigate.rs`): the node, the marks, the correlation bindings and the arrived edges, and the capsa of the earlier text is `Thread<GraphExt>`. The walk's frontier carries one attachment per path, so a predicate pass borrows it and clones nothing.

A `Position` is a number the thread and the pipeline carry and compare; only the graph reads it as a node.

The host. The pipeline evaluates commands and operands against a `Host` (`pipeline/mod.rs`), the trait that holds every question the pipeline cannot answer from a value alone: the root attachment, a property of a node, the provenance at a position, a node's scalar, its rendering as JSON, XML or a table, the navigation of a step, the edges of a node. Every adapter is a host (`graph/host.rs` implements the trait for any `AstAdapter`), and so is the calculator's `Calc`, whose attachment is the unit and whose node questions are refused by name. The generic operand evaluator (`pipeline/operand.rs`) runs on any host; the graph keeps the node-reading half (`graph/operand.rs`) and wraps the generic one with a witness slot for correlation. The registry tags every command with the layer its answer comes from (`Layer::Value` or `Layer::Node`); the calculator's entry refuses a pipeline that names a node command before it runs, as it refuses a head that is not bare.

The parser's reflector. The parser expands macros through an `Expander` (`syntax/expand.rs`), a trait with the usage questions a macro body needs answered; the graph's `Reflector` implements it over an optional adapter, so the syntax layer never names the evaluator. The ambient state a run needs below the pipeline (the collation, the sentence bonds, the refusal channel the library reports into) lives in `value/ambient.rs` as a stack entered per run; the run options the graph adds (the plan, the budget, the reference targets) stay in `graph/run.rs`.

2.3 The Execution Ladder

Database adapters evaluate on a three-rung ladder; every rung is *output-identical* to the one below it (differential tests hold this), so the rungs trade only latency and transfer:

1. **Full pushdown** (`quarb_sql::pushdown`, the exporter's strict mode): the whole query compiles to SQL when every construct is in the verified-safe set — WHERE-filtered selections, whole-table aggregates, witness joins. Strictness is the point: anything whose SQL semantics are not provably identical (LIKE case folding, truthiness, GROUP BY/DISTINCT order, collations, unordered LIMIT) refuses. Row order is part of the contract — the plan names the result table, and each driver appends `ORDER BY` its key from the catalog. Column types are part of it too: Quarb compares by value (numeric-looking text meets a number as a number, a number and other text never meet), SQL by the column's type, and SQLite and MySQL coerce a mixed comparison by rules of their own. A numeric-looking *text* literal therefore never pushes; for

the rest the plan records what it assumes (`ColumnTypes`: the columns it compares with a number, those it compares with text), and qua asks the driver's catalog (`column_catalog`: SQLite's affinity, MySQL's `data_type`) before using a plan, full or partial; an unmet assumption scans, and `--explain` names the column. The strictly typed engines reject the mismatch themselves, which falls back the same way. What remains outside: a SQLite cell that violates its column's declared type. Measured: COUNT over 500k PostgreSQL rows, 2.57 s scanned vs. 0.018 s pushed.

2. **Partial pushdown** (`quarb_sql::partial_pushdown`): the leading run of strictly-translatable row predicates becomes the table's fetch filter (each driver's lazy fetcher accepts a per-table WHERE), and the *original* query runs unchanged over the reduced set — the engine re-applies the pushed predicates, a no-op on pre-filtered rows, so there is no query rewriting. Gates: leading run only (a positional predicate before an expression one sees unfiltered rows), a single `/table/*` branch with no correlations, the table reached exactly once, no crosslink/resolution axes, no `:::/::::` metadata (each would observe the filtering). Since the JSON-column round (2026-07-31), refused predicates in the leading run degrade to *superset prefilters* instead of aborting the plan: fixed-path JSON comparisons emit per-dialect forms proven never to drop a matching row (SQLite typed extract with a `json_type='text'` escape hatch, PostgreSQL `jsonb_typeof` CASE with `::float8`; existence and `!=` floor at `json_type IS NOT NULL`) — extras are re-checked by the re-run, and a server-rejected prefilter surfaces through an eager prefetch and falls back to the scan rather than answering empty. Full JSON string-equality pushdown carries the dialect's validity guard (`json_valid` and kin) and refuses numeric-looking literals (Quarb's `value_eq` coerces those numerically; no extractor matches that). Measured: unpushable grouping over a 1%-selective filter, 2.31 s vs. 0.081 s; JSON numeric compare over 200k rows, 1.46 s scanned vs. 0.25 s prefiltered.

3. **Lazy scan**: catalog-eager, rows-lazy per table (Section on `quarb-relational`); correct for everything, always the fallback.

`qua --explain` prints the decision chain; `qua --no-pushdown` pins the scan rung; `qua --save FILE [--as NAME]` materializes any result as a first-class input (`.db` table / `.json` array) so expensive reductions run once and iteration proceeds locally. For metered engines (a future BigQuery adapter bills by bytes scanned), the ladder's transfer reductions are cost reductions.

2.4 Current Query Subset

The engine implements essentially the whole language. A query is a root-anchored sequence of hops, each an axis followed by a name:

Form	Meaning
`/name`, `//name`	**[live]** child / descendant named `name`
`\name`, `\\name`	**[live]** parent / ancestor named `name`
`>name`, `<name`	**[live]** next / previous sibling
`>>name`, `<<name`, `>>?`, `>>!`, `<<?`, `<<!`	**[live]** sibling reach: all following / preceding siblings at any distance, with proximal / distal quantifiers (matcher before reach, mirroring `//?`); covers CSS `~` and XPath following/preceding-sibling
`//?`, `//!`, `\\?`, `\\!`	**[live]** proximal (nearest) / distal (farthest) reach
`^`, `name$`	**[live]** root anchor / leaf anchor
`*`, `*.rs`, `~(re)`	**[live]** wildcard, glob, and regex name matching
`<code>`, `<a|b>`, `<a><b>`	**[live]** trait filters (alternation, conjunction)
`::`, `:::key`, `::::key`	**[live]** property, core-metadata, and adapter-metadata projection
`[::::size > 1K]`, `[/child]`, `[n]`, `[-1]`, `[2..5]`	**[live]** predicates: comparisons (`= != < <= > >=`), substring `*=`, regex `=~`/`!~`, structural conditions, `and`/`or`/`!`/`not`, positional index and range (negative counts from the end)
`::price * ::qty`, `7 div 2`, `idiv`, `mod`	**[live]** value expressions: arithmetic over numeric readings, in predicates, pipeline commands (`| ::a * ::b`), and pushes (`| .total(::a * ::b)` — a computed column); null propagation, exact integers, overflow to float
`//a || //b`, `| count`, `| sort | join(",")`	**[live]** union, and pipelines over the standard library's command set
`->target`, `<-target`, `->*`	**[live]** crosslink navigation (filesystem: symlinks)

`(/ul/li)+`, `(/p|/div)`, `(->mgr){1,3}!`, `/{2}`	**[live]** path patterns: groups, subpath alternation (strict and tolerated forms), quantifiers (`+ * {n} {m,n} {m,}`) with proximal/distal suffixes, the pattern dot wildcard, and the bare-operator sugar. Simple-path expansion (no node revisited within one path, the anchor included), capped by the adapter's quantifier bound (default 32; `--quantifier-bound` overrides)
`| upper`, `@| count`, `| .n`, `| $.n`	**[live]** per-capsa transform, whole-context aggregation, register push and recall
`@| sort_by(::age)`, `@| top(3, ::fare)`, `@| [..n]`, `| [::age > 30]`	**[live]** keyed aggregates (`sort_by`, `unique_by`, `min_by`/`max_by`, `top`/`bottom`; capsa-preserving, missing keys do not compete), whole-context positional selection, and per-capsa filters
`@| group(::k) | top(2, ::v) @| ungroup`, `$ord`	**[live]** value-keyed grouping (list topics + member capsae, keys as named regs), per-group keyed commands on the plain pipe, `ungroup` flattening, and the `$ordinal` operand (position in the current context, scope-threaded into filters)
`| %.`, `qua q.quarb`	**[live]** the register record view (named regs as a record: the row-wise table view, subsuming the planned `@.|`) and reflection: `.quarb` files open as the query's own arbor (`quarb::reflect`; vocabulary locked v1, enforced by the conformance corpus in `tests/vocabulary.rs`, version on the root as `::::vocabulary`)
`def &f($x): ...;`, `&f(::age)`, `--defs`, `--expand`	**[live]** fragments: parse-time expansion with call-by-name `$param` forms, defs files (`#` line comments stripped in `parse_defs`), and the unparser (`quarb::unparse`) rendering any parsed query back to canonical text (`--expand` = macroexpand); correlation chains and inline-pipe splices as bodies; `rec`/`group` field-name collisions are parse errors (`validate_record_convention`)

```
| .n(cc),@|
.total(count) @| %.,
@| %(total = count;
unique = unique @|
count),sort(::name;
--locale=ru)
```

[live] subcontext bodies: every parenthesis not after a command name is a pipeline fed by its control operator; a headless first command applies to the body's seed (`Operand::Seed`), a bare word names a command (unknown refused by name), a command's own parentheses hold operands then one option segment (`Arg::Opt`, shell spelling, `-=` the void option; undeclared options refused against the registry's `options`); the aggregate push files once on every capsa and passes the context on, `@| %(…)` and `@| %.` read the context as a whole, every reduction keeps the regulae its inputs share (`shared_register`); reflection kinds `cmd` (né `func`), `agg-push`, `agg-record`, `agg-recall`, `seed`, `option`

```
qua '…| sh:tr(a-z;
A-Z) @| go:sort(-r)'
go=exec: /go/bin
sh=shell:
```

[live] environment commands (`quarb::env`): an environment is a namespace of commands the host mounts beside the built-ins, `ALIAS=SCHEME:TARGET` from the command line only (never a model file, a defs file or query text; `--allow-shell` is `sh=shell:` under its older name); the kinds `exec:` (a directory of programs or one program) and `shell:` (the user's shell) with the text framing (`|`: once per capsa, the topic's text on stdin, stdout minus one newline the value, a failed call null; `@|`: once over the context, one line per capsa on stdin, scalars as text and records or lists as their JSON line, every stdout line a capsa carrying the shared regulae, a failed call a refusal); operands as words in order, options first in shell form, the void option onto nothing; an unknown name refused with the alias (`Environment::resolve`; the shell's 127); the gate (`exec::gate`) refuses an unmounted alias before the run; reflection `cmd`/`agg` carry the alias as `lib` (spelled `env` in 0.36.0); `python:` is named and refused until the native framing lands

`| morph:lemma, | morph:parse(-n=1), @| morph:lemma, cargo build -p qua --features morph`	**[live]** command libraries (`quarb::library`): the `CommandLibrary` trait (prefix, declared commands, call, run-time requirements and their availability), a `Synopsis` per command (operands in order with accepted values, options with short and long names and whether a value follows, the pipe or pipes, the input slot by position or option name, the return), checked by `exec::gate` before the run with the synopsis quoted in every refusal; the input slot filled by the pipe unless `$_` places it; options as keyword arguments; after `@|` the stream is the value and a returned list is one capsa per element with the shared regulae; a failed call null after `|` (text on stderr once) and a refusal after `@|`; a node crosses as the record of its name, text and properties. The pilot `quarb-lib-morph` (pymorphy3 through pyo3 `auto-initialize`, feature `morph` on qua, a Python-free default) with `quarb-pyvalue`, the value mapping shared with `quarb-py`; linked libraries are listed in `RunOptions::libraries`, and a library's prefix refuses an environment mount of the same alias

`@| scipy:mannwhitneyu, @| scipy:linkage(--pairs), @| sklearn:pca(-n=2), --features scipy,sklearn`	**[live]** the stream as a table (`quarb-lib-scipy`, `quarb-lib-sklearn`): the glue reads a list of numbers as a sample, a list of records as a matrix (numeric readings per value as the columns in record order, the first text field the row label carried back on every returned row), a list of lists as rows; a k-sample test splits one label and one value per record into samples in the order the labels first appear; `--pairs` reads two text fields and one numeric by position as row, column, entry; a hole (a column missing in one record, a value with no numeric reading, a null) refuses by name unless `--nan-policy=omit`; every test returns `%(statistic; pvalue; …)` and the other fields keep scipy's and scikit-learn's names (`df`, `dof`, `rvalue`, `n_components`); `--digits` rounds to significant digits; the shared call plumbing (`quarb_pyvalue::PyGlue`: the module compiled once, the slot filled, options as keyword arguments with `-` spelled `_`, a Python exception a failed call) also carries `quarb-lib-morph`

```
@| pd:describe |
pd:round(2),@|
np:percentile(90),@|
sp:stats.norm.cdf,
sklearn:cluster.KMeans.fit_predict(3),
| @,| %,@| pivot(row;
column; value)
```

[live] reflected commands and library values: a library that reflects its package (`quarb_pyvalue::Reflective`: `quarb-lib-numpy` `np:`, `quarb-lib-pandas` `pd:`, and the undeclared names of `sp:` and `sklearn:`) resolves a dotted name through the package when the gate first meets it (`CommandLibrary::resolve`), the synopsis built from `inspect.signature` and leaked for the process: required positionals are operands, defaulted parameters options, `*args` the operand `...`, `**kwargs` the option `*` that opens the gate; a method's `self` is the input; a class on the path is constructed with the options its constructor declares, an intermediate object (`groupby`, `rolling`) takes its method, an attribute of a result is read during the run. The input is constructed once (a frame with the first text field as index, or the matrix of numeric fields) and a result that is not plain data is `Value::Foreign` (the `Foreign` trait: kind, display, rows, columns; `PyForeign` holds the object), equal only to itself, handed back whole to the next library command, printed by its own `str()`, exported by its rows in every output form and by the stdlib commands `@` (rows) and `%` (columns), refused by kind elsewhere; `transpose`, `pivot`, `melt` are the stdlib's reshapes

```
= #(@(name; age);
%("Braund"; 22)),
#(%(1; 2); %(3; 4)),
@| #, | :age,
$_:::shape, fit((^/*
| ::label))
```

[live] the table kind and the operand rule: `Value::Table` (`quarb::Table`: an optional header and rows of one width, `null` the hole; a frame with a header, an array without) written `#(...)` (`Token::Hash`, `Operand::Table`, the laws of the literal in `exec::operand::table_literal`), printed as its literal, read by `@`/`rows` (each row as the table has it) and `%`/`columns` (a record of lists, keyed by the header or by position), made by `#` (the table reading of a stream, `stdlib::table_reading`), a row by `| [n]` and a slice by `| [a..b]` (the plain pipe's positional bracket now parses and refuses at the run on anything but a table), `header` and `dim` beside them, a column by the value accessor (`:2` lexes for the array face), printed on the result line with each row on its own line, exported by its rows, crossing into a library command as its rows; `Operand::Index` (`v[2]`, `t[2; "age"]`, ranges in either position; `stdlib::index_value` behind it, `| [n]` and `cell(i; j)`), `row-to-header` and `header-to-row`; `pretty` the renderer (a `rendering` registry spec: a closing aggregate the tool takes off, and a scalar command after a value; `stdlib::pretty_value`, `tabulate_table`; `table` and `--table` aliases through 0.37); the trait `tabula` on every substrate's table (the text model's lowered table, renamed from `table`; HTML and XML `table` elements) and `exec::tabula::node_table` behind `| #` on such a node (lemma-keyed frame for the text model, declared header rows and spans for a grid); a DataFrame or a two-dimensional ndarray coming back is a table (the index the first column unless positional, NaN/NA a hole), a one-dimensional array or a positional Series a list (`quarb_pyvalue::foreign::as_table`). An operand of a library command keeps its kind: every positional argument of a reflected call goes through the same construction as the input, the input fills the first parameter named as data (`InputSlot`, `OperandOrSelf` for a class-rooted path whose instance on the pipe is the receiver), a bare constructor without a data parameter is built from its options alone

`| [::x =~ /(\w+)/] | %(k = $1), quai`	**[live]** match captures (filter-command `=~` binds groups onto the capsa) and the interactive session (living-query lines with rollback, session defs, `:show`/`:undo`; retired from `qua` in favor of the standalone `quai` crate; `NAME=TARGET` mount aliases via `quarb_session::MountSpec` and in-session `:mount` via `Session::set_executor` — in-process mode only; the Python `Session`/kernel accrete defs-only cells through the `parse_defs` binding)
`@| window(-2..0) | mean, @| window(..0) | sum, @| shift(1, ::k)`	**[live]** window functions: offset-range spans over context neighbors as members (partial edges; `window(n)` = trailing count; open start = expanding), `shift` lag with own registers kept, and optional partition keys (per-group windows in original row order)
`"${::name} (${::age})", macro &m($p, @rest): body;`	**[live]** string interpolation (double-quoted `${expr}` holes, evaluated in the current scope; null splices empty) and procedural macros: the body runs at expansion time against the invocation's expansion arbor (argument forms as reflected subtrees, `::form` = each form's text), literal args bind by value, `@rest` collects the remainder, and the text results — joined, reparsed against strictly-earlier fragments — become the expansion (`--expand` = macroexpand)
`macro &pivot!($col): /data/row | $col @| unique | ...`	**[live]** data-aware macros: `!` on the definition mounts the queried dataset as `/data` in the expansion arbor, and every invocation spells the `!` (enforced both ways); outside a hole a parameter is the argument form (call-by-name), inside a hole its text — so a pivot reads distinct data values at expansion and generates a column per value; `--expand` with an input expands against it
`| .(//*.rs @| count), | .t(^/row @| count)`	**[live]** subcontexts (grouped aggregation); a `^`-anchored branch navigates from the arbor root, so a subcontext can consult the whole table (share-of-total columns)

`.c(^/row[::k = $$.k] @\| count), \| %($$1/name:: ; /amt::)`	**[live]** correlated subqueries: `$$` steps a capsa-scope operand out to the invoking capsa (`$$.name`, `$$_`, `$$ord`; the node form `$$::prop` / `$$/child::x` re-anchors on the invoking capsa's node; stacking `$$$` steps further) — wide crosstabs, groupwise transforms; and the correlation keeps its *witness*, so `$*k` in pipeline commands projects the joined side into output (`SELECT a.x, b.y`); the witness travels into every path a command's operand or pipe hop navigates, so a nested step predicate (`(^/c/*[::b_id = $$1::id])`) reads the same thread's binding rather than re-searching the trace (2026-08-18)
`A <=> B[/k:: = $$/k:: and …]`	**[live]** correlation, *driver-first* (2026-07-25): the first expression drives (SQL's `FROM`); each joined expression carries its ON clause in its final step's brackets, reaching the driver as `$$…` and earlier entries as `$*k`; `<=>?` marks the expression it precedes nullable (`LEFT JOIN`); one tuple binds the whole predicate set, searched depth-first with per-entry pruning, and `$*N` may navigate from its context node (`$$1/field::`), so joins work where keys are children, e.g. JSON. Misplaced correlated references are parse errors, never silent nulls
`::ref~>`, `::'$ref'~>/city`	**[live]** cross-reference resolution (JSON: follow a `$ref` JSON Pointer)
`<--(::ref)`, `<--(/scope/*::ref)`	**[live]** reverse resolution (2026-10-05): every node whose `ref` resolves to this one — a declared reference answers from the adapter's index, a scope bounds the search, and the whole-arbor walk runs only where the adapter opts in (`walks_for_references`), else the query is refused; `::a=>(/scope/*::b)` is the value match

`$.r:::provenance`, `:::%provenance`, `:::@origin`, `--reproducible`	**[live]** value provenance (2026-09-06/07): every value carries its *origins* (node, command; `Origins` bounded at eight, the rest counted) and the provenance entries of the nodes it was read from, kept in lockstep with record fields and list items (`Prov` in `pipeline/prov.rs`, thread-local read frames). The value-side `:::` rung (`Operand::ValueMeta`) answers `provenance` (kaiv text), the shaped keys `%provenance` / `@provenance` (record / list: source, path, instant, instant-from, dpid), `origin` / `%origin` / `@origin` (node ids with the command), `path`, `elided`. `resolved_provenance` fills the name-path within the source and completes the instant ladder node → source (mount target mtime) → access (invocation instant); `--reproducible` holds the two lower rungs back. `qua --kaiv` exports per field and per plural on the semicolon with `;+N` elision (`--kaiv-origins`)
`X:.r`, `X:@.`, `X:%.`, `X:(*.)`	**[live]** cross-capsa reads (2026-09-07): the register of the capsa *standing at* the node `X` reaches, read from the command's input snapshot (`Operand::PeerReg`, a lazy `Peers` index by node); location, not lineage — after a fork nothing stands at the fork node, its children do; null outside a pipeline context. What lets one push read its neighbours: `<-link:%.` inside an expression body is the peer's whole register
`((!7))`, `((!$.v:::origin))`	**[live]** the node-id anchor (2026-09-07): `Anchor::Id` takes an expression; a number, a `%origin` record (by `id`), a list forks. `AstAdapter::has_node` guards it (default: probe parent/name under `catch_unwind`, since most adapters index a Vec by id). Ids must be deterministic across runs — minted from the node's identity in the source, never from walk order

`| .v(body){20}, {1;}, {m;n}`	**[live]** the quantified push (2026-09-07): `Command::Repeat` wraps a push-family command (push, expression push, subcontext, record and fields push); each round is one `apply_command` over the whole context, so a round reads the previous round's snapshot; `{N}` exact, `{m;n}` / `{m;}` at least `m` then stop at a fixpoint (no node, topic, or `%.` view changed) or at `n` / the quantifier bound. PageRank over quarb.org is one such push; the stored column matches to 2×10^{-17}

The pipeline runs on *capsae* — the register system's unit, a node paired with a register (its breadcrumbs) and a current topic. A per-capsa `|` transforms each topic; `@|` reduces the whole context; `| .name` pushes the topic onto the register and `| $.name` recalls it; a subcontext `| .(expr)` evaluates `expr` from each node, reduces it to a scalar, and pushes it — grouped aggregation without leaving the query. A push quantified `{N}` repeats over the whole context, each round seeing the last (the iteration loop); `X:.r` reads the register of whichever capsa stands at the node `X` reaches, from the same snapshot; and every value remembers where it was read, so `:::` on a value answers its provenance the way `:::` on a node does.

Any other operator is lexed and rejected with a `not yet supported` error naming the offending construct, rather than silently mis-parsed. Everything in Section 3.4 falls into this category today.

Chapter 3

The Adapter Surface

An *adapter* maps a data source onto the Quarb arbor model — a tree backbone whose edges carry *names*, enriched with non-hierarchical crosslinks. The engine drives all navigation through the `AstAdapter` trait, so this trait *is* the contract between the language and the world.

3.1 The Live Trait

The methods the skeleton actually uses. Signatures are the real ones from `quarb/src/graph/adapter.rs`:

```rust
/// An opaque handle to a node, minted and interpreted by the
/// adapter that produced it.
pub struct NodeId(pub u64);

pub trait AstAdapter {
    /// The root node -- the initial navigation context.
    fn root(&self) -> NodeId;

    /// The tree children of `node`, in document order.
    fn children(&self, node: NodeId) -> Vec<NodeId>;

    /// The name of `node` -- the label of its incoming tree edge.
    /// `None` when the adapter leaves a node unnamed (e.g. the
    /// filesystem root).
    fn name(&self, node: NodeId) -> Option<String>;

    /// The parent of `node`, or `None` for the root.
    fn parent(&self, _node: NodeId) -> Option<NodeId> { None }

    // --- projection surface (defaulted; implement what fits) ---

    /// Adapter-defined classifications, for `<trait>` filters.
    fn traits(&self, _node: NodeId) -> Vec<String> { Vec::new() }

    /// A named property `::prop`.
```

```
    fn property(&self, _node: NodeId, _name: &str) -> Option<Value> {
    None }

    /// The default projection, bare `::` (filesystem: file content).
    fn default_value(&self, _node: NodeId) -> Option<Value> { None }

    /// Adapter metadata `::::key` (filesystem: size, modified, …).
    fn metadata(&self, _node: NodeId, _key: &str) -> Option<Value> {
    None }

    /// Outgoing / incoming crosslinks as (label, node) pairs, for
    /// `->` / `<-` (filesystem: symlinks).
    fn links(&self, _node: NodeId) -> Vec<(String, NodeId)> { Vec::new
    () }
    fn backlinks(&self, _node: NodeId) -> Vec<(String, NodeId)> { Vec::
    new() }

    // --- the reference surface (defaulted) ---

    /// (The name glued to a resolution arrow is its reference label;
    /// the parameter keeps its historical name `hint`; reflection
    /// reports it as `label`.)
    /// Resolve a cross-reference: `::prop-->label` maps a property
    /// value to its in-document target (html: a `#fragment` href
    /// to the element with that id).
    fn resolve(&self, _n: NodeId, _prop: &str, _hint: Option<&str>)
        -> Option<NodeId> { None }

    /// The external reference a property holds -- an
    /// adapter-defined identifier of a document outside the arbor
    /// (html: the absolute URL, relative hrefs joined against the
    /// document's declared URL). On an in-document miss the engine
    /// consults the mount layer's reference targets (identifier ->
    /// mounted root) and lands, or records the identifier among
    /// the run's unresolved external references for the host's
    /// acquisition loop. The engine performs no IO, ever.
    /// (document, location within): the adapter reads both out of
    /// its own reference syntax; the engine splits nothing.
    fn external_ref(&self, _n: NodeId, _prop: &str, _hint: Option<&str
    >)
        -> Option<ExternalRef> { None }

    /// The element `fragment` names inside `node`'s document --
    /// the landing rung of a fragment-carrying external reference.
    fn resolve_fragment(&self, _n: NodeId, _frag: &str)
        -> Option<NodeId> { None }

    /// The property carrying the node's own reference, for the
    /// bare arrow `//a-->` (html: an anchor's href); and the
    /// relation an edge carries (html: `rel`), the label falling
    /// back to the property name.
    fn ref_property(&self, _n: NodeId) -> Option<String> { None }
    fn ref_label(&self, _n: NodeId, _prop: &str) -> Option<String> {
    None }

    // --- the index hooks (defaulted; an indexed store answers) ---
```

```
    /// The document an external identifier names (a URL -> its
    /// page), the arrow's landing rung after the mount layer.
    fn document_by_ref(&self, _id: &str) -> Option<NodeId> { None }
    /// `<--(::prop)` from an index (a declared reference);
    /// `None` means no declaration answers.
    fn reverse_resolve(&self, _n: NodeId, _prop: Option<&str>,
        _hint: Option<&str>) -> Option<Vec<NodeId>> { None }
    /// Whether an unscoped `<--` may walk the whole arbor here
    /// (the in-memory formats say yes; a catalog says no).
    fn walks_for_references(&self) -> bool { false }
    /// `index SCOPE::prop;` (a model statement): the scoped nodes
    /// holding `value`, and those whose property resolves to a
    /// target, from an index the model keeps. Cost only: the
    /// caller still tests each node, so the answer is the walk's.
    fn index_lookup(&self, _scope: &str, _prop: &str, _v: &Value)
        -> Option<Vec<NodeId>> { None }
    fn index_referrers(&self, _scope: &str, _prop: &str,
        _label: Option<&str>, _target: NodeId)
        -> Option<Vec<NodeId>> { None }
    /// Can a reference *declared* on `source` (a catalog key, a
    /// model `ref`, a reference property the format defines) land
    /// in `target`'s container? Schema, never data; `None` where
    /// nothing is declared. Decides `<--(SCOPE::*)` with no label.
    fn can_refer_to(&self, source: NodeId, _target: NodeId)
        -> Option<bool> { self.ref_property(source).map(|_| true) }
    /// The reference properties declared on `source` into
    /// `target`'s container: what `t<--` and `t<--(SCOPE)` read.
    fn reference_properties(&self, source: NodeId, _target: NodeId)
        -> Vec<String> { self.ref_property(source).into_iter().collect
    () }
    /// Whether the source resolves references at all (a CSV file
    /// does not): both arrows are refused where it does not.
    fn resolves_references(&self) -> bool { true }
    /// The label a label-less reverse arrow also resolves under,
    /// when the question is about `target` (a database: its
    /// table). None by default; the engine derives none.
    fn reverse_label(&self, _target: NodeId) -> Option<String>
        { None }
    /// The trees other trees refer to by name: (identity, node);
    /// an identity ending in a separator is a base. Declared in a
    /// model (`tree IDENTITY: PATH;`), automatic for a mount's
    /// own target. The engine reads prefix and separator, no more.
    fn tree_identities(&self) -> Vec<(String, NodeId)> { vec![] }
    /// Whether `::prop-->` needs a reference label here (a column
    /// with no key, a property graph): refused without one.
    fn resolve_needs_label(&self, _n: NodeId, _prop: &str) -> bool
        { false }
    /// A node's property names, for `<--(SCOPE::*)`; an element
    /// lists its attributes.
    fn property_names(&self, _n: NodeId) -> Vec<String> { /* … */ }
    /// `//name` below `node` from an index, with each hit's depth;
    /// `None` means walk. The answer must equal the walk's.
    fn descendants_named(&self, _n: NodeId, _name: &str)
        -> Option<Vec<(NodeId, usize)>> { None }
    /// The search fast path: one step's plan (axis, name, the
    /// equality conjuncts the engine lowered, a limit, an order)
    /// answered from an index -- `None` walks. The default folds
```

```
    /// `children_named` / `descendants_named` / `descendants_where`.
    fn select(&self, _from: NodeId, _plan: &StepPlan<'_>)
        -> Option<Selection> { ... }
    /// The dry-run form of `select`: which conjuncts the adapter
    /// would apply, from its catalog alone (`scope` is the literal
    /// path from the root to the step's base); `None` cannot say.
    fn select_plan(&self, _scope: &[&str], _plan: &StepPlan<'_>)
        -> Option<Vec<bool>> { None }
    /// What reaching the step would cost, from the catalog: row and
    /// byte intervals, requests, a latency class, a confidence
    /// (a dry run, statistics, a file's size, an arena's count).
    fn estimate(&self, _scope: &[&str], _plan: &StepPlan<'_>)
        -> Option<Estimate> { None }
    /// How many nodes the step reaches, without making them:
    /// `@| count` right after a bare step asks (SQL's COUNT(*), an
    /// index bucket's length). `None` walks.
    fn count(&self, _from: NodeId, _plan: &StepPlan<'_>)
        -> Option<usize> { None }
    /// A batching hint before the engine reads the properties
    /// `names` of every node in `nodes` (the next command projects
    /// them): the projection twin of `prefetch_links`.
    fn prefetch_properties(&self, _nodes: &[NodeId], _names: &[&str])
    {}
    /// A batching hint before a link sweep over `nodes`.
    fn prefetch_links(&self, _nodes: &[NodeId], _dir: LinkDir) {}

    // --- identity and provenance (defaulted) ---

    /// Whether `node` is a node of this arbor -- the guard of the
    /// node-id anchor `((!expr))`. Default: probe parent and name
    /// under catch_unwind.
    fn has_node(&self, node: NodeId) -> bool { ... }
    /// What the source recorded about `node`: source, path within
    /// it, instant (and which rung it came from), dpid; a list when
    /// several sources stand behind one node (mounts, grafts,
    /// models layer entry by entry).
    fn provenance(&self, _n: NodeId) -> Provenance { ... }
    fn provenance_list(&self, n: NodeId) -> ProvenanceList { ... }
}
```

The navigation methods are required; the projection methods carry defaults, so an adapter implements only what its domain offers. Core metadata (`:::key` — index, depth, counts, `is-leaf`, ...) is computed by the engine from the navigation surface, so no adapter implements it.

Three observations that the filesystem work has already forced:

- **Names live on edges, not nodes.** `name` returns the label of a node's *incoming* edge — its role in its parent — exactly as the specification's edge-name model requires. For the filesystem this is simply the filename.
- **`NodeId` is opaque and adapter-private.** The engine never inspects the `u64`; it only passes ids back to the adapter. Each adapter chooses its own identity scheme (the filesystem adapter uses an interning arena index).

- **New trait methods need their forwarders.** The run path wraps adapters in layers (the decoding `Sourced` layer), the session in `Dyn`, mounts in `MountAdapter`/`Shared`, models in `ModelAdapter` – a defaulted method silently answers `None` through any wrapper that does not forward it. The plain wrappers (those that override at most a method or two) write their forwarding with `quarb::forward_ast_adapter!(FIELD; except m, …)`, whose method table is checked against the trait by a test in `adapter.rs`, and a probe adapter is run through each shipped wrapper – so a new trait method is forwarded by all of them once it is added to the table. The composing wrappers (`MountAdapter`, `ModelAdapter`, the graft, the reflection views) still forward by hand: when adding to the trait, sweep those. The case in point (2026-09-07): `Dyn` had never forwarded `answers_to`, so a web level behind the session's boxed door answered `//page` with nothing – its rows answer to their kind through that method. The sweep of 2026-10-06 found three hooks no wrapper forwarded (`descendants_where`, `has_node`, `provenance_list`), and `ModelAdapter` dropping `link_property` and aliased metadata, so an edge property or `::::size` answered nothing under `--model`; the macro round of 2026-10-07 found the Python binding's `Dyn` forwarding 29 methods of 45. A model's mount targets stay relative to its own file (the tool resolves them once, against the file it was given), so a `used` model's targets are carried over by the `use` path as written.

- **Integers compare as integers.** Two integers are compared, keyed for joins and `unique`, and sorted exactly; the float reading enters only when one side is not an integer (`SortKey::Line` carries the integer beside the float, and -0.0 keys as 0.0). Above 2^{53} the float reading alone would merge neighbours.

- **The core is smaller than the crate.** The temporal fragment (instants, durations, periods, time expressions, the calendar names) is the `quarb-time` crate, re-exported by `quarb` at its old paths (`quarb::temporal`, `period`, `timex`). Three features, on by default, carry what one function needs: `decode-formats` (the structured readings of `decode`: serde_json, serde_yaml_ng, toml, quick-xml), `kaiv` (the kaiv output of `| kaiv` / `@| kaiv`), `desm` (syndesmos as the session's sentence bonds). Off, the function refuses with the feature's name. The sentence bonds themselves are a trait (`quarb::SentenceBonds`): the engine segments through whatever the tool set, syndesmos or another.

- **The plan is read before the run.** `Query::plan(&adapter)` (`quarb::plan`) reads off the syntax tree, per branch and per step, the axis the fast path covers, the literal name, the equality conjuncts lowered, the literal path to the step's base where every step before it was a bare child hop by name, and what the adapter says from its catalog: `select_plan` (which conjuncts it would apply) and `estimate` (nodes, bytes, requests,

latency, confidence), composed along a branch (`Estimate::then`) and across branches (`beside`), unknown where no estimator answers. `Plan::roots` names the first child of every anchored branch – under a mount table, the mounts the query touches; an unanchored search is every mount. A `Run` may carry a `Budget` (nodes, bytes, requests; `strict` refuses an unknown estimate): a query the estimate puts beyond it is refused before anything runs, and the `Outcome` carries the plan. `Plan::to_kaiv` emits it as a `quarb/plan` payload (`qua --plan` prints it and stops; `qua --budget nodes=N,bytes=N`). `Model::reachable_roots(nam` (quarb-model) closes a set of mounts over the `ref`, `rel` and `tree` declarations, for a replay plan. The relational model takes a driver's *estimator* (`RelationalModel::with_estimator`: the table and the plan's column equalities in, an `Estimate` out, asked once per run per table and equalities): BigQuery prices the statement with a dry run (bytes exact, a bound for a clustered table) and reads the row count from `tables.get`; Postgres, MySQL and Oracle ask their planners (`EXPLAIN`, `EXPLAIN FORMAT=JSON`, `EXPLAIN PLAN`); SQL Server reads `sys.partitions`; DuckDB its catalog's `estimated_size`; Athena the table's crawler properties; SQLite `sqlite_stat1` after `ANALYZE`, unknown without it. The SQL adapters forward the whole trait through `forward_ast_adapter!`, so `select`, `count`, `select_plan` and `estimate` reach the model.

- **A mount may be latent.** The mount adapter declares a mount by name, target and opener (`MountAdapter::declare`) and opens it the first time a query reaches it: a hop by its name, a listing of the root's children (so an unanchored `//x` opens every mount before the scope memo is built), a reference landing on a declared identity. `tree_identities` answers a latent mount from its declaration with a stand-in root id that opens the mount when touched; the every-mount hints read the active set. A failed opener is a refusal and an empty mount. A live table takes new mounts (`push`) and identities (`declare_identity`). A declaration may carry a `catalog` closure: what the target holds before it is opened, so the plan prices a step from the mount's root while the mount is latent.

- **A catalog prices a source before it is read.** `quarb::Catalog` is substrate-neutral — entries and bytes as a range, a latency, a confidence — and every substrate states one cheaply: `quarb_archive::catalog` from a zip's central directory or a tar's headers, `quarb_open::catalog` from a bounded directory listing or a file's size, a warehouse from its dry run. `Estimate::from_catalog` turns it into a root step's estimate as bounds (entries are files or members, the estimate counts nodes: a child step reaches at most the entries, an unfiltered descendant step at least every entry), the catalog is a lookup by literal path (`CatalogLookup`: a folder's own listing, one member's header), so `Estimate::at_catalog` narrows the price to the path the plan records

— a bare hop by name is one entry, the reading step under it is priced by that entry's catalog, a path the catalog cannot place (a node inside a document) is a bounded unknown under the whole; a mount table prices a root step across its mounts; a database file is left to its store's estimator once open, and a hop by name under a table is a row by key, at most one. The plan lowers the leading predicate as a comparison (`Conjunct::Compare`: any operator, any literal; a text equality stays `Equals`, the index's) and the step's trait filters, and asks the catalog with them (`Ask`): a listing applies `::size` and `<file>`/`<dir>` before counting and says so (`Catalog::applied`), a store's statistics price an equality on an indexed column (SQLite: `sqlite_stat1`'s rows per key); what a catalog cannot apply leaves the whole count as the bound; `quarb::Unopened` is an adapter that is only its catalog, for planning and budgeting without opening (`qua --plan` / `--budget` price every path input this way first). Once open, an in-memory adapter answers `estimate` through `adapter::arena_estimate`, a counted walk of its own nodes (JSON, CSV, XML, HTML, text, fs, archive), bounded past two million nodes. An upper end of `u64::MAX` means unbounded: a lenient budget passes it, a strict one refuses. The budget's cap (`Budget::cap`, carried as `StepPlan::cap` and handed to every catalog producer) short-circuits the pricing itself: a listing, a tar header walk or an arena count stops at the first entry past the budget and reports a lower bound that already exceeds it, so pricing a directory of a hundred thousand files under `nodes=1000` costs a thousand reads and one. Pricing is opt-in: a run with neither `plan` nor a budget never asks an adapter for an estimate and never reads a catalog, so the default path runs the query straight, at no planning cost.

- **The conformance suite is the contract's test.** `quarb::conformance::check(&adapter, budget)` walks a sample of an adapter's nodes and names every pair the engine relies on that the sample breaks: a child's `parent`, `has_node`, `answers_to` its own name, `children_named` and a child `select` finding it, `has_trait` over `traits`, `property_names` over `properties`, `provenance_list`'s first entry, a descendant `select` against the walk. An adapter's tests call it on a fixture and assert the list is empty (the JSON and HTML adapters do); an index that answers short is caught there, not by a query.

- **The shell gate is checked twice.** The static walk refuses a query that names `sh(...)` or a backtick command anywhere it can stand (record fields, list items and field bases included); the executor refuses the command again when it runs, so a path the walk does not know cannot reach the shell without `--allow-shell`.

- **A child's standard input is fed from a thread with SIGPIPE blocked.** The shell command and the environment commands write the topic (or the context's lines) to the child from a thread of their own, so the

parent drains the child's output meanwhile (`env::feed_stdin`). A child that stops reading early — `head`, a program that fails — closes the pipe under the writer; with the signal blocked on that thread the write fails with EPIPE and the call takes its proper outcome (null, or a refusal over the context), where the default disposition `qua` restores for its own stdout would kill the whole process silently (exit 141). Found by feeding `@| u:head(-n=3)` a novel.

- **What a run is given is the run's, never the adapter's.** The invocation instant `now()` denotes, the shell gate, the quantifier bound override, reproducible provenance, the reference targets, the output collation and the sentence bonds travel in a `quarb::Run`; a tool parses a `Query` once (data-aware macros expand under the same `Run`) and runs it as often as it likes, each run answering an `Outcome` (the result, the closing table taken off for rendering, the unresolved references). The executor reads the options from the run in progress, and a run begun from inside another – an adapter answering through the engine, a model's constructor queries – inherits them, so nothing rides on an adapter method or a global. The adapter keeps only `quantifier_bound`, the depth its own structures need (metatheca's chain), which a run may override. The convenience entries (`quarb::run` and its `_with_defs` / `_traced` twins) run under the ambient options: the instant the tool pinned, the bonds it set, the gate closed.
- **The root may be unnamed.** `name` returns `Option` precisely so an adapter can leave the root without a name, as the filesystem root `/` does.
- **A failed fetch is a refusal, never an empty answer.** The trait has no error channel below the entry points, so a read that fails inside an evaluation — a table load, a label listing, a relationship fetch, a count, a paginated listing cut off mid-way — records a refusal (`quarb::record_refusal`; `quarb::fetched` / `fetched_opt` wrap a `Result` that way) and the query errors with the adapter's message. A listing that failed is left uncached, so the next query reads again. The one legitimate emptiness is the server's own word — LDAP's `noSuchObject` and `insufficientAccessRights`.
- **Bounds where data set the size.** Every bound refuses with a message naming it: nesting (`MAX_NEST`, carried into holes, expansions and generated text), `repeat` / `indent` (`MAX_REPEAT_BYTES`, 64 MiB), outline levels (`MAX_OUTLINE_DEPTH`, 64 below the shallowest), unit exponents (`MAX_EXPONENT`, 32), mounts (`MAX_MOUNTS`, 254: `MountAdapter::try_new`), what an archive may decompress into memory (`MAX_ARCHIVE_BYTES`, 1 GiB, an xz stream checked against its index first). The executor's tree walks run on an explicit stack, so a source's depth is never the thread's.

3.2 The Source Boundary

Text is text inside the language; bytes meet it at two boundaries, both the engine's, in `quarb::source`. An adapter decides nothing about encodings.

- **Decoding** (`source::decode`, carried by a `SourceDecoding` per mount): `?encoding=LABEL` on the mount, then a byte-order mark, then the declaration the format's reporter finds in the bytes, then strict UTF-8 — a source that reaches the last step and is not valid UTF-8 is refused, naming the option. Nothing is guessed from bytes and nothing is decoded lossily; within a decided encoding a malformed sequence becomes U+FFFD and is counted. Labels are the Encoding Standard's (`encoding_rs`); the CJK tables ride the `cjk` feature of `quarb` (on for `qua`, off for the playground and the extension).
- **Reporters** are plain functions over bytes in the adapter crates: `quarb_xml::declared_encoding` (the prologue; the reader itself is `quarb::source::xml_declared_encoding`, shared with the packages), `quarb_html::declared_encoding` (the meta charset prescan), `quarb_maildir::declared_encoding` (the MIME charset of a message's own headers). A host chooses the reporter by the source's name (`quarb_compose::declared_encoding`); the adapter never decodes.
- **Layers that read bytes** take the rule and answer for what they decoded: `Sourced` (the wrapper `qua` puts on a mount it read itself) answers `::::encoding` and `::::replaced` on the root; `ComposeAdapter` (`with_decoding`) reads a grafted leaf through its path and answers on that leaf; `ArchiveAdapter` (`open_with_decoding`) on each member it reads, tar and zip names included (an entry name is bytes under the rule, never the format's CP437 guess; zip members are read by index, so a decoded name need not match the library's); `MaildirAdapter` / `ImapAdapter` per message; `quarb_web::fs::open_dir_with_decoding` per page; `GitAdapter` / `MetathecaAdapter` (`open_with_decoding`) on a blob (one that decodes under none has no text; git's own structured output must be UTF-8, paths unquoted and messages transcoded by `--encoding=UTF-8`); the packages — docx, odt, epub — on each XML member through `parse_with_decoding` (the option, a byte-order mark, the member's XML prologue, else strict UTF-8; a member that decodes under none is an error naming it, never a part silently missing); `TreeSitterAdapter` / `CodeModel::open_with_decoding` and `quarb_text::read_conllu_dir_with` on a source file. A response body over HTTP is read whole under `source::text_from_reader` (a 256 MiB bound, strict UTF-8 — the wire formats are UTF-8 by their standards), never cut at a client library's cap nor replaced.
- **Delimited text** is decided the same way, never sniffed: `CsvAdapter::parse_declared` takes `?delimiter=` off the mount (`CsvOptions`), else an Excel `sep=X`

first line, else the extension's default; a header without the decided delimiter but another candidate throughout is refused naming the option. Numbers in text read in the canonical form (`quarb::number`: digits, sign, `.`, exponent; grouped thousands under the default convention); a source that writes decimal commas declares `?decimal=comma` and its cells with a reading arrive in canonical form, so every reading downstream agrees. `| number(LOCALE)` reads text under a locale's convention from a small embedded table; `@| csv(";"; …)` takes the delimiter as a quoted first operand (the parser carries it as `delimiter=;`).

- **Dates in a locale's words**: `strptime(fmt; LOCALE)` / `tp` and `strftime(fmt; LOCALE)` / `tfmt` read and write the `%B %b %A %a` names from `quarb::datenames`, a small embedded CLDR table (by size of language: en, zh, es, ar, pt, ru, ja, de, fr, tr, it) listing every form a locale writes (the date form `октября`, the stand-alone `октябрь`, the abbreviations); every form parses, case folded, longest first, and writing takes the date form where the format carries a day (`%d`, `%e`) and the stand-alone form where it does not (`%B %Y`: октябрь 1899). Format literals match as written (`'%d %B %Y г.'`). Without a locale the English names; a locale with no table refuses. The temporal reading of bare text stays ISO-8601.
- **Periods and time expressions** (spec: Time Expressions and the Timeline). `Value::Period` is a start second and a grain (`quarb::period`: year to second, ISO-8601 at reduced precision in print and in the interchange forms). The *calendar reading* (`Value::calendar_reading`) reads a period, reduced or full ISO text, an integer as a year, an instant at its printing grain. `value_eq` / `value_cmp` test for a period operand first and compare as intervals: nested periods are equal, disjoint ones ordered. A period has its own sort key (start, then the coarser first), no equality key (a search takes the scan), no numeric reading (arithmetic is null), and reads as its first instant wherever a command goes through the temporal reading. `period([GRAIN])` and `grain` are scalar commands. `timex(LOCALE)` (`quarb::timex`) is an expanding scalar command like `words`: one record per expression. Recognition is a list of templates per locale (`timexdata.rs`, the eleven locales of the date names), each a regular expression over slots (numeral, unit, month, weekday, day, year, clock, part of day) compiled once per locale. The earliest match wins, then the longest, then the rule listed first, and an overlapping match is dropped. Widening a list is the whole of teaching the reader an expression. No stemmer and no tagger takes part. Digits of other scripts, the typographic apostrophe and the Turkish dotted capital are folded one character for one before matching, so `at` counts characters of the text as written. A word boundary closes every template in the spaced locales, so an abbreviation's final point (`p.m.`, `г.`) stays with the sentence. `time_anchor(T)` and `@| timeline` share `period::anchor`: a span

moves by calendar units keeping the anchor's grain (the month's last day where the target month lacks the day), except that an anchor finer than the day drops to the day when the offset names no unit finer than the day, a weekday offset and a time of day ask for a day or finer, a date with fields left out takes them from the anchor (under `forward`, the first such date at or after it). `timeline_command` is row-aligned like `shift`: it evaluates the expression and the anchor per capsa, folds the point over the context (a change of anchor value starts a chain), and adds `when` and `from` to each record. A value that is not a `timex` record is carried as `%(value; when; from)`.

- **Unit libraries**: `units LIBRARY;` in a model imports a kaiv `.faiv` library (path relative to the model file; `use` merges) into the unit table the unital reading consults for every source the model mounts: `ModelAdapter::unit_scale` answers from the imported definitions first (`kaiv::unit::scale_with`), then the source's own. Loaded at construction, refused by `check`: an unreadable or malformed file, the same file twice, a name the table holds already (built in, or an earlier library). Unit names are letters in any script (the unital tokenizer admits Unicode letters), matched byte for byte as kaiv lists them; aliases read as their primary. A quantity prints its unit as written. Through kaiv-rs dev (0.15.0, path dependency) until kaiv publishes.

- **The output locale** (`quarb::locale`) governs rendering only, never the value: a tool decides the locale in force by its layers, nearest first — `--locale`, `QUARB_LOCALE`, `~/.quarb/config.toml` (`[output] locale`), the system's `LC_ALL` / `LANG` (`C` and `POSIX` canonical) — and renders what it prints through `OutputLocale`: numbers' decimal mark and grouping (CLDR's, with its minimum grouping digits), instants in the locale's medium date and time pattern through `strftime_in`, `@| csv`'s delimiter (the semicolon where the decimal mark is the comma), and `sort`'s collation when the query names none (`set_default_collation`, read by `default_text_order`; text alone is collated, numbers and instants keep their own order). Digits are grouped only when asked (`--grouping`, `QUARB_GROUPING=1`, `[output] grouping = true`): a year is not `1,876`. The engine itself never renders under a locale. A command is a value, the same wherever the query runs, so `@| csv` and `@| table` inside a pipeline are canonical; when one *closes* the query the tool renders it, as it renders `--csv` and `--table`: `Query::for_rendering` runs the query with the closing command replaced by an internal stand-in that yields the rows, and the `Outcome` carries the command's columns and named delimiter (`ClosingTable`). `@| json`, `@| jsonl` and `@| kaiv` stay canonical. A reader (`number`, `tp`) takes its locale from its operand alone, and refuses a locale it has no table for. `qua --explain` reports the locale and its layer; the gates run with the environment's locale cleared. The session and wasm doors have no lo-

cale yet (the browser's `navigator.language` would be their system layer).

- **Encoding** (`source::encode`): `@| encoding(LABEL)` is the output boundary and stands last. The engine has no byte value, so the tool takes the command off the end of the query, runs the rest and encodes what it would have printed (qua's `OutSink`); met anywhere else it is refused by the executor. The command before it decides what a character the encoding cannot hold becomes: after `@| json` / `@| jsonl` the format's `\u` escape (`source::Escape::Json`); after anything else the output is encoded while representable and refused at the first such character. Mail headers' RFC 2047 encoded words decode under their own charset through the same tables (`quarb-maildir`).

`bibref(STANDARD)` (`quarb::bibref`) reads one reference text under a citation standard — `gost`, `gb7714`, `cmos`, `ams`, an open tag set the command refuses outside of, naming it — into a record of the fields the grammar places without doubt (English names: `author`, `title`, `year`, `journal`, `pages`, ...), its `genus` and the list `label`; a text that is no reference is null, as `strptime` answers for a text that is no date. Each grammar is a hand-written reader of the standard's punctuation (GOST's areas at `. –`, GB's type code after the title, Chicago's sentences and quoted titles, AMS's comma fields), the numbering prefix stripped first. The same module finds citations in prose (bracketed numbers and ranges with a pinpoint, author-year parentheses) and `TextModel::lower_references` (quarb-text) applies both for the `?references=STANDARD` mount option: a paragraph or list item that parses becomes a `bib` node under the canonical campi (with the English aliases attached when no census is), its `::onym` the written number, the list ordinal, or party and year; the citations become `cit` nodes and the resolution pass runs again.

3.3 Worked Example: the Filesystem Adapter

`quarb-fs` implements the live trait over a directory tree. It materializes the tree once, at construction, into arrays indexed by `NodeId` (path, children, parent); the trait methods are then trivial lookups:

```
impl AstAdapter for FsAdapter {
    fn root(&self) -> NodeId { self.root }
    fn children(&self, node: NodeId) -> Vec<NodeId> { self.children[
    node].clone() }
    fn name(&self, node: NodeId) -> Option<String> {
        if node == self.root { return None; }   // "/" is unnamed
        self.paths[node].file_name().map(|n| n.to_string_lossy().into()
    )
    }
    fn parent(&self, node: NodeId) -> Option<NodeId> { self.parents[
    node] }
}
```

Hidden entries and directory links [live]. An entry whose name starts with a dot carries the `<hidden>` trait (the mount root never does); such entries are in the tree only under `--hidden`. A symlink is a node with a `target` crosslink, and a symlink to a directory of the tree is also transparent to an explicit child step, as it is in the file system: with `latest -> drafts`, `/latest/a.md` is the node `/drafts/a.md`, the same one `/latest->*/a.md` reaches. The trait method behind this is `child_host(node)`, the node whose children a child step reads; it defaults to the node itself, and the wrapping adapters (mounts, grafts, models, reflection) forward it. The descendant walk reads `children` alone, so `//` sees each file once and a cycle of links cannot trap it. A broken link, and a link whose target lies outside the mounted tree, stay leaves.

Ignore rules [live]. The materializing walk uses `ignore::WalkBuilder`—the same machinery ripgrep uses—so by default the adapter respects `.gitignore`/`.ignore` files and skips hidden entries. This came directly from running the skeleton: `qua '//*.rs'` in the engine's own tree first returned a build artifact under `target/`, because nothing yet excluded it. The adapter now honors ignore rules by default, and `qua` exposes `--hidden` and `--no-ignore` to opt back in. The walk is currently eager (a targeted query still walks the whole tree); a lazy traversal with cascading ignore rules is a planned optimization. This is the pattern the guide follows: a concrete problem, surfaced by running the code, drives the next slice.

3.4 The Planned Surface

The rest of the arbor model, specified in the language but not yet built. Each becomes a live trait method (or a related capability) as its engine slice lands. The intended shapes:

Capability	Query	Adapter surface (sketch)
Pattern search	`=>`	`search(pattern) -> Vec<NodeId>`.

These signatures are provisional. They are recorded here so the direction is legible, not to freeze a contract—the filesystem adapter (and the JSON adapter after it) will reshape them as they are implemented.

Chapter 4

Building and Running

```
# from the engine workspace root
cargo build            # build the whole workspace
cargo test             # unit + integration + doc tests

# query a directory tree
cargo run -p qua -- '//*.rs' src/     # all Rust files under src/
cargo run -p qua -- '/README.md'      # a single top-level file

# query a document
cargo run -p qua -- '/users/0/name::' data.json
cargo run -p qua -- '//book[::pages > 200]::::id' catalog.xml
cargo run -p qua -- '//a::href' page.html
```

`qua <QUERY> [PATH]` runs `QUERY` against the directory `PATH` (default: the current directory) and prints each matching path on its own line, so it composes in shell pipelines alongside `sort`, `uniq`, and friends.

Chapter 5

Planned Integration Paths

Beyond the direct `AstAdapter` implementation shown above, these integration paths are described in the specification but **[planned]** (none is implemented yet):

- **Petgraph adapter.** A ready-made adapter over a `petgraph::Graph`, for Rust callers who transform their AST into a graph once and query it at speed. See `petgraph-adapter.md` in this project for the design note.
- **Python bindings.** A `quarb` Python package exposing the engine and an `AstAdapter` base class.
- **C FFI.** A C ABI for embedding the engine in other languages.

Each will be added once the core language surface is substantial enough to be worth binding.